

# FinanceNarrate AI

## Executive Financial Insight Engine

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Generative AI Course

Google Gemini 2.5

FastAPI + Pandas

Chart.js Dashboard

# The Problem with Financial Reporting

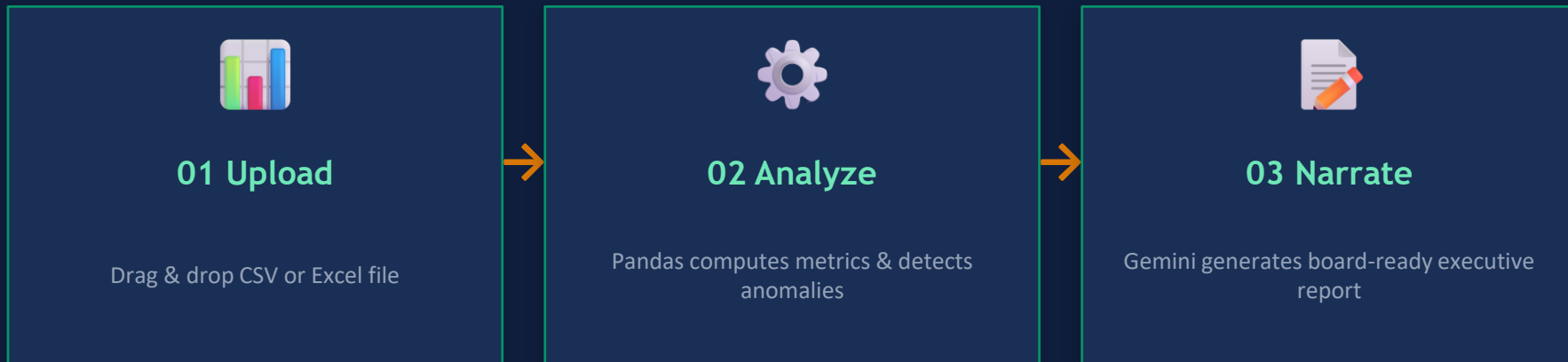
## WITHOUT AI

- ✗ 4–6 hours/month writing reports manually
- ✗ Error-prone, inconsistent output
- ✗ Raw spreadsheets must be re-interpreted each cycle
- ✗ Insights stale before reaching decision-makers
- ✗ No resources for automation in SMBs

## WITH FINANCENARRATE AI

- ✓ Full report generated in under 10 seconds
- ✓ Consistent, structured AI-generated narrative
- ✓ Pre-computed metrics feed directly into LLM
- ✓ Instant insights — always current
- ✓ One web app — zero setup required

# What is FinanceNarrate AI?



- ◆ Combines Pandas analytics with Google Gemini GenAI
- ◆ Original, context-aware text — never templated
- ◆ Browser-based — no installation required
- ◆ No manual writing needed at any stage

# How We Does This Project



## Google Gemini 2.5 Flash

State-of-the-art LLM  
generating original financial  
narrative from computed  
metrics context

**AI sees metrics — not raw data**

### Prompt Engineering

Carefully crafted instructions to control output format, tone, and length

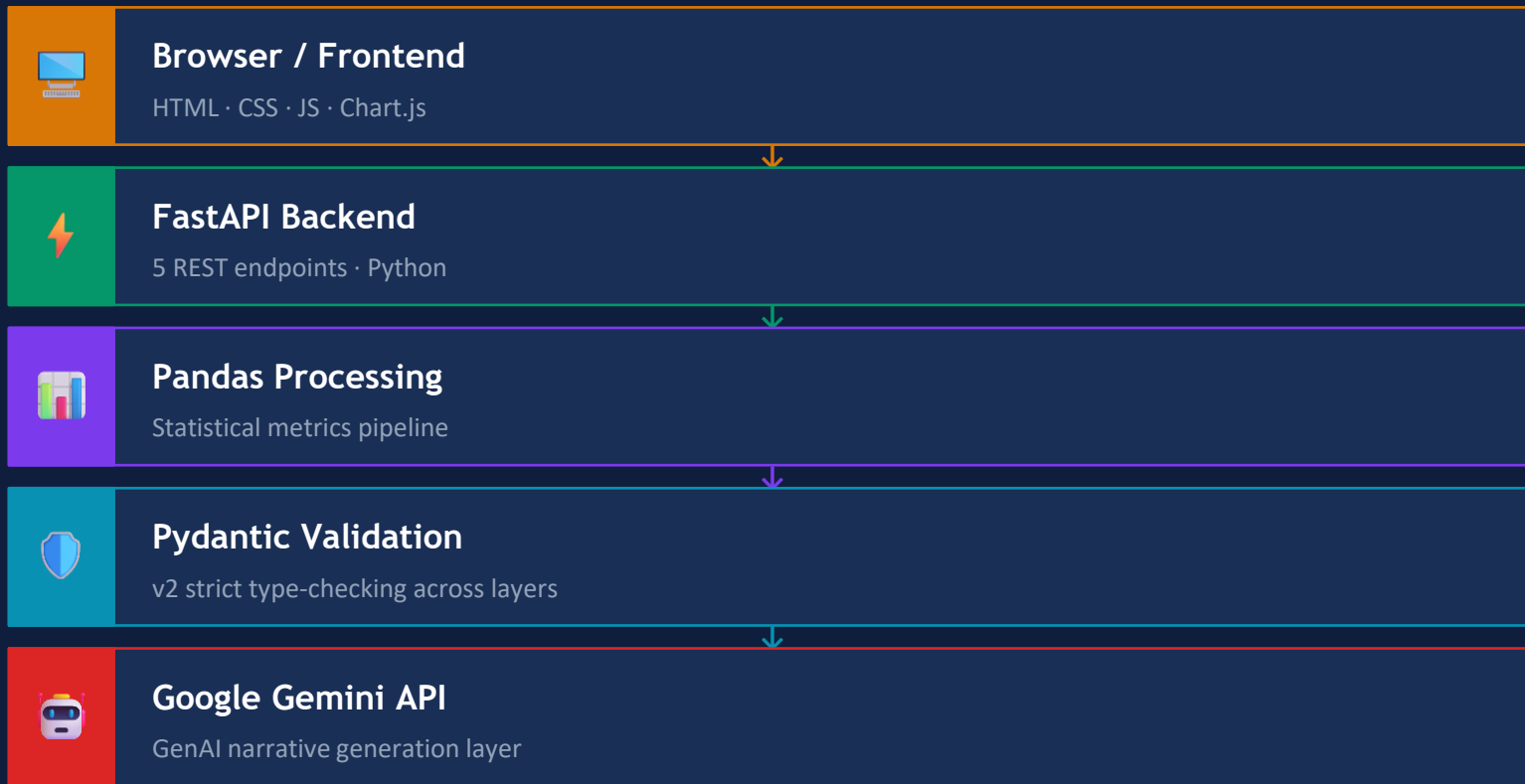
### Structured Output Generation

Forces Gemini to return strict JSON schema — 4 predictable sections every time

### Context-Augmented Generation

Injects pre-computed metrics as context for specific, accurate, hallucination-free responses

# Technical Architecture



*No database required — in-memory state management with Python dicts*

# What the System Does



## File Upload

Drag-and-drop CSV/Excel with automatic column validation



## Revenue Aggregation

Monthly totals + month-over-month growth calculation



## Anomaly Detection

Z-score ( $2\sigma$  threshold) flags unusual expense spikes



## Dip Detection

Flags revenue months with >15% decline automatically



## AI Narrative

4-section Gemini report: Summary · Trends · Anomalies · Actions



## Export & Share

One-click copy or download report as .txt file

# Prompt Engineering – Bridge to Generative AI

*Designing LLM inputs to produce reliable, structured, domain-specific outputs every time*

## Role Assignment

`"You are a senior financial analyst preparing a board-ready report"`

## Context Injection

`Full computed metrics JSON embedded directly in the prompt body`

## Format Constraint

`"Return ONLY valid JSON with exactly these four keys: summary, trends, anomalies, recommendations"`

## Length Constraint

`"Write a 3-4 sentence summary. Provide exactly 2-3 action recommendations."`

*Result: consistent, parseable, hallucination-resistant financial narratives on every run*

# Development Methodology



## Spec-Driven Development

Requirements → Design → Tasks defined before any code was written



## Pipeline Architecture

Sequential independent stages — Validator → Processor → LLM Client → API → Frontend



## Statistical Methods

Z-score ( $2\sigma$ ) for expense anomaly detection · 15% threshold for revenue dip detection



## Property-Based Testing

Hypothesis library generates hundreds of random inputs — revenue round-trip & MoM growth verified



## Separation of Concerns

Each layer — Validator, Processor, LLM Client, API, Frontend — isolated and independently testable



# Results & Demo

51

Automated Tests  
34 unit + 17 API — all passing

<10s

Full Pipeline  
Upload → Analyze → Narrative → Export

4.5 $\sigma$

Anomaly Detected  
Operations, November — \$95,000

- 📦 Sample dataset: 24 rows · 12 months · 5 expense categories
- 🚨 1 expense anomaly correctly detected (Operations, Nov 2023 — z-score 4.5)
- 📉 1 revenue dip detected (November 2023: -33.3% from October)
- ⚡ Gemini generated complete 4-section executive report in under 5 seconds
- 🔗 GitHub: [github.com/durganuhya02/finance-narrate-ai](https://github.com/durganuhya02/finance-narrate-ai)

# CONCLUSION & FUTURE SCOPE

## FinanceNarrate AI

*Generative AI · 2025*

### Key Takeaways

- ✓ Successfully automates knowledge work with Generative AI
- ✓ Combines deterministic Pandas statistics + Gemini narrative
- ✓ Hours of manual work reduced to seconds
- ✓ LLMs most effective with structured pre-processed context

### Future Scope

- 📅 Auto-email monthly reports to stakeholders
- 📁 Multi-file quarter/year comparison reports
- 🔒 User auth + persistent report history (DB)
- 📊 QuickBooks / Tally integration
- 🧠 Fine-tuned financial LLM for deeper domain expertise